VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

MILESTONE 1

Project Initialization, Design Process & Core Setup

Week 1 & 2

Milestone Progress Document

1. Project Overview

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System is an AI-powered manufacturing quality-inspection platform intended to detect product defects from images, identify quality issues, classify defect types, and provide production analytics. The platform is designed to reduce manual inspection effort, improve product quality, minimize manufacturing defects, and increase production efficiency.

2. Milestone 1 Objective

Establish the foundation of the VisionInspect AI platform by defining the quality-inspection workflow, designing the system architecture and database schema, preparing development environments, implementing authentication and role-based access, integrating the MVTec AD dataset, and creating the initial image-upload and inspection workflow.

3. Scope of Work

- Defined project objectives and the manufacturing quality-inspection workflow.
- Planned the overall system architecture and database schema.
- Prepared frontend and backend development environments.
- Implemented authentication and role-based access for quality engineer and factory supervisor accounts.
- Integrated the MVTec AD dataset as the planned dataset source.
- Built the product-image upload workflow and initial inspection dashboard.

4. Planned System Modules

- User Management
- Authentication and authorization
- Role management

5. Key Deliverables

- Project initialization and architecture design
- Database schema and workflow plan
- Frontend and backend project setup
- Authentication and authorization foundation
- Dataset integration plan
- Image upload and inspection dashboard foundation

6. Expected / Recorded Outcomes

- Understanding of AI applications in Industry 4.0 and smart manufacturing.
- Understanding of system architecture and database design.

- Initialized frontend and backend components.
- Working foundation for authentication and image inspection management.

7. Technology Stack

Backend: Python (FastAPI) | Frontend: React.js / Next.js | Database: PostgreSQL / MongoDB

AI & Computer Vision: OpenCV, TensorFlow, PyTorch, YOLO, CNN models, Scikit-learn, NumPy and Pandas.

Deployment & Development: Docker, Docker Compose, AWS/Azure, Git/GitHub, VS Code and Postman.

8. Performance & Evaluation Focus

The project specification identifies defect-detection accuracy, precision, recall, F1-score, mAP, inspection automation rate, defect identification accuracy, false-defect detection rate, image-processing speed, inspection response time, and concurrent inspection capability as key metrics.

9. Challenges / Notes

This milestone document is based on the supplied VisionInspect AI project specification. Where implementation evidence, measured model metrics, screenshots, deployment URLs, or issue logs were not included in that source, this document does not invent those details. Actual measured results and screenshots can be added to the GitHub copy before submission.

10. Next Milestone / Continuation

Implement image preprocessing, image-quality analysis, defect-detection model training, prediction workflows, and inspection monitoring.

Conclusion

Milestone 1 establishes the planned progression of the VisionInspect AI system toward an end-to-end manufacturing defect detection and quality inspection platform. The work in this milestone provides the foundation for the next stage of implementation and final demonstration.